Teeth Reconstruction and Performance Capture Using a Phone Camera

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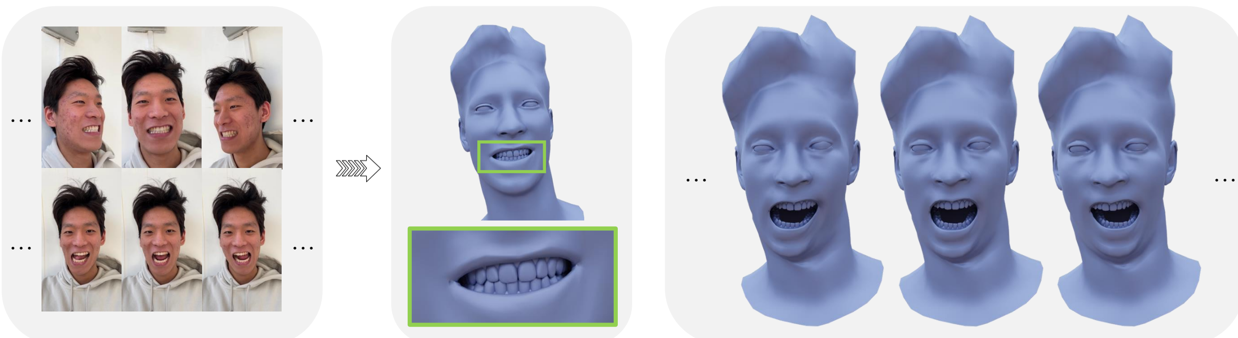

Figure 1. Given a monocular video, our method performs personalized dental geometry reconstruction and teeth-inclusive facial performance capture.

Abstract

We present the first method for personalized dental shape reconstruction and teeth-inclusive facial performance capture using only a single phone camera. Our approach democratizes high-quality facial avatars through a non-invasive, low-cost setup by addressing the ill-posed monocular capture problem with an analysis-by-synthesis approach. We introduce a representation adaptation technique that maintains both mesh and SDF representations of teeth, enabling efficient differentiable rendering while preventing teeth-lip interpenetration. To overcome alignment challenges with similar-appearing dental components, we leverage foundation models for semantic teeth segmentation and design specialized optimization objectives. Our method addresses the challenging occlusions of teeth during facial performance through optimization strategies that leverage facial structural priors, while our semantic mask rendering loss with optimal transport-based matching ensures convergence despite significant variations in initial positioning.

1. Introduction

3D modeling and performance capture of real human faces is an essential step toward immersing ourselves in the digital world and has significant applications in virtual and augmented reality, human telepresence, and beyond. Recent advancements have improved the quality of face modeling and performance capture, paving the way for broader adoption in facial avatars. However, faces contain components with largely varying geometry, appearance, and motion patterns. To further enhance quality and enable more fine-grained controllability, many recent works have focused on modeling the specific facial components, such as hair [66, 67], eyes [36, 45], mustache [7] and neck [62], rather than taking them as a whole.

Among these facial components, teeth have received relatively less attention, but they play a critical role in face avatars. Teeth are fixed to the skull and jaw and move accordingly, which can help regularize jaw motions. Teeth influence airflow during pronunciation and thus influence various visemes, so the teeth movement impacts the quality of speech animations and is a key feature for lipreading [38, 40]. Teeth exhibit significant individual variation in shape, number, and alignment, making them an essential feature for identity representation [4, 17, 52, 56]. Furthermore, teeth serve as rigid obstacles to constrain the motion of lips, affecting the final lip movements. Despite their importance, existing shape reconstruction methods [21] predominantly focus on facial skin regions, often omitting teeth in their results, leading to an uncanny valley effect. While a recent neural rendering approach [26] models a complete face including teeth, its volumetric representation poses challenges to exporting the teeth shape, limiting their effectiveness for downstream applications.

There are a few previous works that have attempted to address the problem of teeth reconstruction and performance capture from RGB cameras. The work proposed by Wu et al. [55] reconstructs static teeth but does not support performance capture for dynamic expressions. A key challenge in performance capture is the complex occlusion relationship between teeth and lips. Although intra-oral reconstruction methods [16, 58] are unaffected by lip occlusion, they rely on specialized capture devices and invasive capture setups, making them unsuitable for performance capture of facial expressions. [43] achieves high-quality tracking of the inner mouth but requires a multi-camera capture rig, limiting its accessibility. To the best of our knowledge, there is no existing method that enables low-cost teeth reconstruction and performance capture from a single daily device.

In this paper, we present the first method for personalized dental geometry reconstruction and teeth-inclusive facial performance capture using only a single phone camera, as shown in Fig. 1. Unlike previous approaches that require complex capture setups, our method democratizes high-quality facial avatars through its non-invasive capture process and accessibility of low-cost equipment. We adopt an analysis-by-synthesis approach [8] to optimize tooth geometry and determine its relative pose with the skull and jaw [31]. This process requires inverse rendering of teeth jointly with a mesh-based facial model and constraining tooth shape and motion. To achieve this, we propose a representation adaptation technique that provides dual representations of teeth to maintain gradient propagation necessary for optimization. Firstly, we leverage a neural SDF-based tooth morphable model [61] to constrain tooth shape optimization, and employ differentiable level set extraction [39] to obtain a mesh representation of teeth for efficient differentiable rasterization alongside the face. Secondly, to constrain the motion of teeth, we maintain the SDF representation to enable a contact loss that prevents interpenetration between teeth and lips, ensuring physically plausible contact between them.

Teeth comprise multiple components with similar appearance, making precise alignment between image data and 3D dental models challenging. To address this, we leverage a recent foundation model for video segmentation [44] to extract semantic masks of teeth, and design optimization objectives specifically suited for geometry and pose refinement based on these masks. Initial teeth positioning can substantially deviate from person-specific dental arrangements due to variations in skull morphology. Conventional differentiable rendering approaches that rely on perpixel losses do not guarantee convergence given this initialization [57]. We overcome this limitation by designing a semantic mask rendering loss that establishes correspondences through optimal transport-based matching, facilitating convergence in shape and pose optimization.

Teeth tracking in facial performance involves distinct occlusion states, like visible, partially visible, and completely occluded by the lips. We observe that conventional differentiable rendering approaches struggle particularly with transitions between visibility and occlusion. While this problem is generally challenging, we leverage the unique structural prior of the human face to make a key observation: teeth can only be occluded by the lips and surrounding facial regions. Based on this insight, we propose two specialized optimization strategies that address the transitions between these two visibility states. Our experimental results demonstrate that these strategies effectively overcome the occlusion challenges and significantly enhance the accuracy of teeth tracking. Code will be released.

Our contributions are as follows:

- We present the first method for personalized teeth reconstruction and performance capture using a single phone camera.
- We present a representation adaptation technique for teeth that leverages both mesh and SDF representations, enabling efficient differentiable rendering while preventing intersections.
- We introduce specialized optimization strategies that leverage facial structural priors to effectively track teeth during challenging transitions between partial visibility and complete occlusion.

2. Related Work

2.1. Facial Performance Capture

Modeling 3D faces from images and videos has been a topic of extensive research, which can be broadly categorized into multi-view and single-view methods. Studio-level multi-view camera setups [6, 9] enable high-quality facial performance capture, but are not widely accessible due to their high cost. Monocular face tracking methods [11, 12, 53] typically rely on 3D Morphable Models (3DMM) [8] to overcome the ill-posed nature of the problem, but their accuracy is limited by the expressiveness of the model.

To improve the accuracy of face shape, some works have introduced additional deformations to 3DMMs [13, 23, 37, 47, 51]. However, these methods only improve the accuracy of the skin region, lacking the modeling of specific facial features like eyes and inner mouth. Other works have employed neural network-based geometric representations to reconstruct complete heads from images or videos, such as volumetric densities [5, 22, 35, 41] and signed distance functions (SDF) [25, 33, 65]. While these methods reconstruct regions beyond the skin, they do not allow control or manipulation of the individual facial components. A series of works have focused on improving the quality of facial performance by modeling specific facial components, such as hair [67], eyes [29, 36, 45], lips [24], and neck [34, 62]. Please refer to surveys [19, 68] for a comprehensive overview. Modeling teeth and incorporating them into facial performance capture is worth exploring to further enhance the quality of facial avatars.

2.2. Teeth Reconstruction

Since the teeth and jaw are anatomically connected, modeling the jaw is beneficial for facial reconstruction that incorporates the teeth. The FLAME model [31] uses jaw parameters to deform the facial skin via linear blend skinning. Several works have focused on modeling the jaw using multi-view cameras and markers to build jaw rigs [60, 69], as well as tracking jaw pose [70]. In addition, a skeleton-consistent 3DMM has been developed from CT data to model the jaw and face in a unified framework [42]. While most of the works on jaw modeling focus on accurately capturing jaw positions, the shape of the teeth is often less emphasized.

Medical research, particularly in orthodontic treatment, has placed greater emphasis on teeth modeling, accumulating extensive 2D and 3D data. To visualize orthodontic treatment predictions, various approaches have been explored, including 2D image synthesis [14, 15, 18, 46, 59], 3D scan animation [20, 49], and treatment previews in augmented reality (AR) [3]. The use of plaster cast 3D scan data in orthodontics has enabled the development of the teeth morphable models [55, 61], which more accurately models the shape variations of teeth. However, integrating these teeth models into facial reconstructions requires solving the challenge of aligning them with the head model.

Early works in 3D teeth reconstruction focused on single tooth modeling [1] and image-based tracking from CT scans [2]. Recent advancements have extended to multi-view RGB image reconstruction, including intra-oral reconstruction [16, 58] and multi-view shape reconstruction [54], as well as appearance capture based on differentiable rendering [48]. However, these methods typically require the use of a dental cheek retractor, making them unsuitable for dynamic facial expression performance capture. More recently, Rasras et al. [43] proposed a high-quality inner mouth tracking method that includes teeth, though it still necessitates a multi-camera capture rig. In contrast, our work introduces the first method for teeth reconstruction and performance capture using a single phone camera.

3. Preliminaries

In this section, we introduce the facial and dental parametric models that serve as the geometry prior of our method.

FLAME. FLAME [32] is a mesh-based face parametric model. It defines a parametric mapping to facial shape through linear interpolation and blend skinning:

$$\mathrm{FLAME}(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}) = W(T_p(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}), \mathbf{J}(\boldsymbol{\beta}), \boldsymbol{\theta}, \mathcal{W}), \quad (1)$$

where $\boldsymbol{\beta} \in \mathbb{R}^{|\boldsymbol{\beta}|}$ represents identity shape parameters, $\boldsymbol{\theta} \in \mathbb{R}^{3k+3}$ encodes pose parameters for four joints (skull (or neck), jaw, and both eyeballs) plus global rotation, and $\boldsymbol{\psi} \in \mathbb{R}^{|\boldsymbol{\psi}|}$ contains expression coefficients. $T_p(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi})$ interpolates the base mesh, which is then transformed by the linear blend skinning function W according to blend weights $\mathcal{W} \in \mathbf{R}^{k\times n}$, joint locations $\mathbf{J} \in \mathbb{R}^{3k}$, and pose parameters $\boldsymbol{\theta}$. While FLAME lacks teeth geometry, its anatomically-based articulated skull and jaw joints provide an ideal foundation for incorporating drivable teeth into the model.

Dental Morphable Model. The Dental Morphable Model (DMM) [61] is a neural SDF-based parameteric model of teeth and gums. It employs a neural network f conditioned on a set $\boldsymbol{z} = \{z_i\}_{i=1,\cdots,m}$ of gum and teeth latent codes (with $m = 30$, two for gums and 28 for teeth). The network maps spatial locations $\mathbf{x}$ to per-component signed distances $\{s_i\}$ and semantic indicators $\{\delta_i\}$, following the definition in [61]. The surface is defined by the zero level set of the aggregated signed distance $\{\mathbf{x} \mid \Phi(\mathbf{x}; \boldsymbol{z}) = 0\}$:

$$\Phi(\mathbf{x}; \boldsymbol{z}) = \sum_{i=1}^{m} w_i(\mathbf{x}) \cdot s_i(\mathbf{x}), \text{with } w_i(\mathbf{x}) = \frac{\delta_i(\mathbf{x})}{\sum_{j=1}^{m} \delta_i(\mathbf{x})} \quad (2)$$

where w_i are blending weights.

The compositional model assembles the full geometry from semantically distinct subcomponents, enabling instance-level rendering of individual teeth.

4. Method

Our method reconstructs personalized teeth shape from a monocular video, integrates it with a face model to create a drivable facial rig, and optimizes expression parameters to accurately match both face and teeth in the input. Sec. 4.1 details how we combine models from Sec. 3 into a coherent facial rig that maintains proper contact while enabling efficient differentiable rendering. Sec. 4.2 explains our optimization framework for teeth shape reconstruction and performance capture from the input video.

4.1. Teeth-Face Composition

4.1.1. Representation Adaptation

For efficient animation and inverse rendering optimization, we unify the mesh-based face and SDF-based teeth into a cohesive representation, while preserving the SDF representation for collision detection. We convert teeth to mesh representation to leverage its compatibility with facial optimization pipelines and the performance advantages of differentiable rasterization. Given a latent code $\boldsymbol{z}$, we represent the teeth SDF as $\Phi(\mathbf{x}; \boldsymbol{z})$ and extract the zero-level

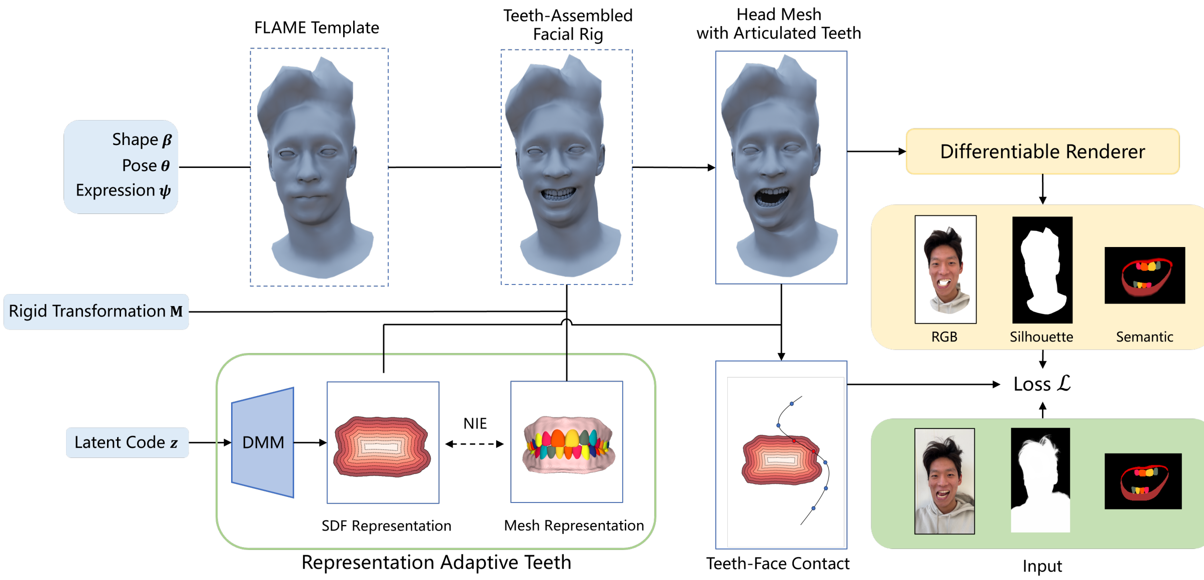

Figure 2. Overview of our method. We employ representation-adaptive teeth (Sec. 4.1.1), where the mesh representation is assembled onto the face to create an animatable rig (Sec. 4.1.2), while the SDF representation is used to regularize and prevent teeth-face penetration (Sec. 4.2.3). Both teeth and face are jointly optimized through differentiable rendering, with specially designed losses and optimization strategies for semantic masks (Sec. 4.2).

set using Marching Cubes to obtain the mesh representation $\mathbf{T}(\boldsymbol{z})$. This mesh is subsequently animated with facial expression and used for differentiable rendering to compute losses. During backpropagation, gradients on $\mathbf{T}(\boldsymbol{z})$ determine vertex deformations, which propagate to the SDF network parameters through NIE [39]. We leverage the SDF representation to prevent teeth-lip intersections, which frequently occur due to their close anatomical relationship. For a lip vertex $\mathbf{x}_{\text{lip}}$ in canonical space, we detect intersections when $\Phi(\mathbf{x}_{\text{lip}}; \boldsymbol{z}) < 0$ and adjust expression and pose parameters accordingly. This dual representation approach harnesses both the geometric constraints from SDFs and the computational efficiency of mesh-based animation and rendering.

4.1.2. Teeth-Assembled Facial Rig

For anatomically meaningful animation, we assemble teeth to the skull and jaw. We divide the teeth mesh into upper and lower components as $\mathbf{T}(\boldsymbol{z}) = \{\mathbf{T}_{\text{up}}(\boldsymbol{z}), \mathbf{T}_{\text{low}}(\boldsymbol{z})\}$. We introduce rigid transformation parameters $\mathbf{M}_{\text{up}}$ and $\mathbf{M}_{\text{low}}$ to align teeth from DMM canonical space to the FLAME template space. Leveraging FLAME's kinematic tree, we extend Eq. (1) to:

$$\begin{aligned} &M(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{z}, \mathbf{M}_{\text{up}}, \mathbf{M}_{\text{low}}) = \\ &W(\{T_p(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi}), \mathbf{M}_{\text{up}}\mathbf{T}_{\text{up}}(\boldsymbol{z}), \mathbf{M}_{\text{low}}\mathbf{T}_{\text{low}}(\boldsymbol{z})\}, \\ &\quad \mathbf{J}(\boldsymbol{\beta}), \boldsymbol{\theta}, \{\mathcal{W}, \mathcal{W}_{\text{up}}, \mathcal{W}_{\text{low}}\}). \end{aligned} \tag{3}$$

The one-hot blend weight matrices $\mathcal{W}_{\text{up}}$ and $\mathcal{W}_{\text{low}}$ rig upper teeth to the skull joint and lower teeth to the jaw joint respectively. This ensures teeth move naturally with skull and jaw movements while remaining independent of lip deformations. Parameters $\boldsymbol{z}$, $\mathbf{M}_{\text{up}}$, and $\mathbf{M}_{\text{low}}$ are jointly optimized.

4.2. Analysis-by-Synthesis Optimization

We jointly optimize the face and teeth using Nvdiffrast's [28] differentiable rasterization framework, denoted as Π. Here we focus on our teeth-specific semantic mask fitting approach, teeth constraints, and strategies for addressing occlusion challenges, while leaving face-specific losses and complete optimization objectives in the implementation details in Sec. 5.1.1.

4.2.1. Semantic Mask Supervision

The visually similar components of teeth make RGB constraints alone insufficient for precise alignment. Leveraging the DMM's per-tooth semantics, we introduce semantic supervision for each individual tooth. We extract semantic masks $\mathcal{S} \in \mathbb{R}^{m_{\text{vis}}}$ for visible teeth using a recent video segmentation foundation model [44], where $m_{\text{vis}} \leq m$. For each vertex $\mathbf{x}_{\text{v}}$ on mesh M, we derive soft semantic labels from the blending weights $\{w_i(\mathbf{x}_{\text{v}})\}$ computed according to Eq. (2), while assigning zero vectors to non-teeth regions. This produces per-vertex semantic labels $\mathbf{S} \in \mathbb{R}^{|M| \times m_{\text{vis}}}$, which we render as

$$\widehat{\mathcal{S}} = \Pi(M, \mathbf{S}), \tag{4}$$

and define the semantic loss as

$$\mathcal{L}_{\text{sem}} = \|\widehat{\mathcal{S}} - \mathcal{S}\|_1. \tag{5}$$

4.2.2. SemanticXY Supervision

Traditional pixel-wise semantic loss requires overlapping regions between target and rendered masks to avoid local minima [28]. However, teeth reconstruction often involves discrepancies between pose initialization and target person-specific teeth positions, leading to non-convergence or inaccurate fitting. Inspired by DROT [57], we employ optimal transport-based matching that propagates gradients from long-range correspondences back to mesh M. We first render screen-space point coordinates $\mathbf{U}(M)$ as an XY coordinate map:

$$\widehat{\mathcal{U}} = \Pi(M, \mathbf{U}(M)). \tag{6}$$

Though visually trivial, this map is parameterized by 3D mesh coordinates, enabling gradient backpropagation.

To establish long-range correspondences between pixels, we concatenate $\mathcal{SU} = [\mathcal{S}, \mathcal{U}] \in \mathbb{R}^{m_{\text{vis}}+2}$, treating them as high-dimensional points for optimal transport-based matching. We denote $\text{Warp}(\widehat{\mathcal{SU}}, \mathcal{SU})$ as the result of mapping $\widehat{\mathcal{SU}}$ according to the correspondences established through this matching process. Based on this mapping, we define our semanticXY loss as:

$$\mathcal{L}_{\text{semXY}} = \ell_{\text{semXY}}(\widehat{\mathcal{SU}}, \mathcal{SU}) = \|\text{Warp}(\widehat{\mathcal{SU}}, \mathcal{SU}) - \widehat{\mathcal{SU}}\|_1 \tag{7}$$

The $\text{Warp}(\widehat{\mathcal{SU}}, \mathcal{SU})$ function both establishes correspondences between semantically similar pixels across potentially distant regions and, due to the influence of $\mathcal{U}$, prefers neighboring pixels when multiple semantic candidates exist. Importantly, the $\mathcal{L}_{\text{semXY}}$ propagates gradients through $\widehat{\mathcal{U}}$ that ultimately flow back to M, enabling effective geometric optimization.

4.2.3. Regularization

Model prior loss. We regularize our DMM to generate plausible teeth using a prior loss on the latent code:

$$\mathcal{L}_{\text{prior, teeth}} = \lambda_{\text{teeth}} \|\boldsymbol{z}\|_2^2. \tag{8}$$

Contact loss. Using the representation adaptation method from Sec. 4.1.1, we prevent intersections between teeth and the face by defining:

$$\mathcal{L}_{\text{contact}} = \| \max(-\Phi(\mathbf{M}_*^{-1} W^{-1}(\mathbf{x}_{\text{v}})), 0)\|_1, \tag{9}$$

where $\mathbf{M}_* \in \{\mathbf{M}_{\text{up}}, \mathbf{M}_{\text{low}}\}$ together with W^{-1} transforms vertex positions $\mathbf{x}_{\text{v}}$ from the facial mesh back to DMM canonical space.

4.2.4. Occlusion-Aware Optimization Stategy

Teeth tracking presents unique challenges compared to face tracking due to frequent occlusion. The most difficult aspect is managing transitions between the visible state and complete occlusion—a scenario where conventional differentiable rendering approaches typically fail. By leveraging the specific occlusion relationship between teeth and lips, where teeth can only be hidden by the lips and surrounding facial regions, we propose two optimization strategies to effectively handle these visibility state transitions.

Face-vanishing strategy. When rendered teeth are completely occluded while the target teeth are visible, gradient propagation paths are blocked, preventing teeth optimization. Since teeth must be occluded by facial regions, we remove these occluding facial elements and render additional semanticXY masks $\widehat{\mathcal{SU}}_{\text{no\_occ}} = \{\widehat{\mathcal{S}}_{\text{no\_occ}}, \widehat{\mathcal{U}}_{\text{no\_occ}}\}$. We then optimize using an additional loss:

$$\mathcal{L}_{\text{f-v}} = \ell_{\text{semXY}}(\widehat{\mathcal{SU}}_{\text{no\_occ}}, \mathcal{SU}). \tag{10}$$

This encourages the teeth to emerge from the occluded state by guiding them toward the visible target.

Teeth-hiding strategy. When rendered teeth are visible but the target teeth are completely occluded, we recognize that the occlusion should occur from specific facial mesh regions. For example, lower teeth should be occluded by the lower lip and surrounding facial area. We identify this region and render its mask as $\widehat{\mathcal{SU}}_{\text{occ}}$. We then impose a loss term to encourage the rendered teeth mask to be contained within this region:

$$\mathcal{L}_{\text{t-h}} = \ell_{\text{semXY}}(\widehat{\mathcal{SU}}, \widehat{\mathcal{SU}}_{\text{occ}}). \tag{11}$$

Although this occluding region is typically larger than the teeth mask itself, the optimal transport-based matching preferentially selects closer candidates, guiding the teeth toward the nearest occluding regions until the appropriate occlusion is achieved.

5. Experiments

In this section, we first describe the datasets and our implementation details. Next, we compare our work with various state-of-the-art approaches from two perspectives: performance capture and shape reconstruction, qualitatively and quantitatively. Finally, we demonstrate our contributions by ablation study. Since our work primarily focuses on geometry rather than appearance, we present geometry results below. Please also refer to the supplementary video for the video version of the experiments and additional results.

Datasets. We use a phone (iPhone 14) to capture RGB videos at 1080p resolution from four participants. Specifically, the capture process is divided into two phases. In phase 1, participants are required to maintain a toothy expression for approximately 20 seconds while we record by moving the phone in an S-shaped trajectory toward them. In phase 2, the camera is positioned directly in front of the participants' faces as they perform various expressions that involve dynamic tooth movements. For data preprocessing, We use the software Agisoft MetaShape to obtain the camera's intrinsic and extrinsic parameters, while also processing video frames to extract landmarks, face parsing maps, and silhouettes. We use SAM2 [44] to obtain the semantic mask of the teeth. Segmenting one tooth requires the user to manually click the tooth in one frame, and then the detected mask will be propagated to the whole video. Specifically, in phase 1, we segment all visible teeth, while in phase 2, to ensure segmentation stability, we only segment the eight incisors that are most prominently visible. In addition, to quantitatively evaluate our results, we make a synthetic dataset using Unreal MetaHuman that replicates our capture setting. The synthetic dataset also contains ground truth depth maps for quantitative evaluation.

5.1. Implementation details

Our method is implemented using PyTorch. For the tracking optimization process, we first optimize all parameters on the frames from phase 1, where the identity-related parameters $\{\boldsymbol{\beta}, \boldsymbol{z}, \mathbf{M}_{\text{up}}, \mathbf{M}_{\text{low}}\}$ are shared across these frames. After obtaining a good initialization, we optimize the parameters in a frame-by-frame fashion using the frames from phase 2. Finally, we perform a global optimization using all frames. Tracking a video of approximately 500 frames using our method takes around 8 hours on an NVIDIA RTX 3090.

5.1.1. Face Tracking

Our monocular performance capture builds upon VHAP [41], extending it with teeth representation that is jointly optimized with the face. We represent facial appearance using a texture map, while teeth appearance is modeled with an appearance MLP F with architecture following [50] defined in the DMM canonical space. This combination produces per-vertex colors $\mathbf{A} \in \mathbb{R}^{|M|\times 3}$ for the teeth-assembled head mesh, which is rendered under SH lighting ζ to generate RGB images $\widehat{\mathcal{I}}$ and head silhouettes $\widehat{\mathcal{M}}$.

For target features, we detect facial landmarks $\mathbf{L}$ using [10][1] and extract target silhouettes $\mathcal{M}$ with a face parsing model [64]. Following [41], we add per-vertex offsets ΔM defined on the FLAME mesh.

We optimize our tracking through multiple loss terms:

$$\mathcal{L}_{\text{rgb}} = |\widehat{\mathcal{I}} - \mathcal{I}|_1, \tag{12}$$

$$\mathcal{L}_{\text{sil}} = |\widehat{\mathcal{M}} - \mathcal{M}|_1, \tag{13}$$

$$\mathcal{L}_{\text{ldmk}} = |\widehat{\mathbf{L}} - \mathbf{L}|_1, \tag{14}$$

Additionally, we incorporate regularization terms for model parameters and mesh offsets:

$$\mathcal{L}_{\text{prior, face}} = \lambda_{\text{shape}}\|\boldsymbol{\beta}\|_1 + \lambda_{\text{exp}}\|\boldsymbol{\psi}\|_1 + \lambda_{\text{pose}}\|\boldsymbol{\theta}\|_1, \tag{15}$$

$$\mathcal{L}_{\text{prior, offset}} = \lambda_{\text{offset}}\|\Delta M\|_1 + \lambda_{\text{lap}}\text{LapSmooth}(\Delta M), \tag{16}$$

[1] https://github.com/hhj1897/face_alignment

where $\text{LapSmooth}(\cdot)$ denotes a Laplacian smoothing term.

Overall, we minimize the full objective defined as:

$$\begin{aligned}\mathcal{L} = \; & \mathcal{L}_{\text{rgb}} + \mathcal{L}_{\text{sil}} + \mathcal{L}_{\text{ldmk}} \\ & + \mathcal{L}_{\text{sem}} + \mathcal{L}_{\text{semXY}} + \mathcal{L}_{\text{contact}} \\ & + \mathcal{L}_{\text{prior, face}} + \mathcal{L}_{\text{prior, offset}} + \mathcal{L}_{\text{prior, teeth}} \\ & + \epsilon_{\text{f-v}}\mathcal{L}_{\text{f-v}} + \epsilon_{\text{t-h}}\mathcal{L}_{\text{t-h}}\end{aligned} \tag{17}$$

where ϵ are binary indicators of strategy employment: $\epsilon_{\text{f-v}} = 1$ only when rendered teeth are completely occluded while target teeth are visible, otherwise 0; $\epsilon_{\text{t-h}} = 1$ only when rendered teeth are visible but target teeth are completely occluded, otherwise 0. We use $\lambda_{\text{shape}} = 0.3$, $\lambda_{\text{exp}} = 0.03$, $\lambda_{\text{pose}} = 0.3$, $\lambda_{\text{offset}} = 30.0$, $\lambda_{\text{lap}} = 60.0$.

5.2. Comparison on performance capture

Our method focuses on the facial performance capture of the teeth. Therefore, we compare with two recent monocular facial tracking methods. VHAP [41] is a mesh-based approach and MonoNPHM [25] is an SDF-based approach. Compared with FLAME [31], both works extend the representation of the mouth interior region.

The qualitative results evaluated on real datasets are shown in Figure 3 and the supplementary video. The mesh region representing the teeth in VHAP fail to align with the target, suggesting that it cannot accurately track the teeth motion. MonoNPHM struggles to produce geometry that accurately represents the teeth, making it difficult to align with the target teeth. Our method, on the other hand, not only generates plausible dental geometries but also provides accurate teeth motion.

For quantitative comparison, we calculate the Chamfer Distance(CD) for each frame in the synthetic dataset as a geometric metric to evaluate the performance capture accuracy. Specifically, we rasterize the mesh to depth map and convert it to point cloud for calculating the geometric metric. Marching Cube are performed to get the mesh for MonoNPHM. We calculate our geometric metric only in the regions where the teeth are visible. The average metrics are shown in Table 1. Our method shows a significant improvement compared to previous approaches.

5.3. Comparison on shape reconstruction

We evaluate the performance of our method in shape reconstruction, particularly in the teeth region, by comparing with TensoSDF [30], a state-of-the-art SDF-based method for shape reconstruction from multi-view images. We evaluate only on the frames from phase 1, during which the facial expressions remain static. Additionally, we further evaluate our method on another multi-view RGB image dataset released by NeRSemble [27].

Qualitative results are shown in Figure 4. Although TensoSDF can reconstruct a complete face including the teeth,

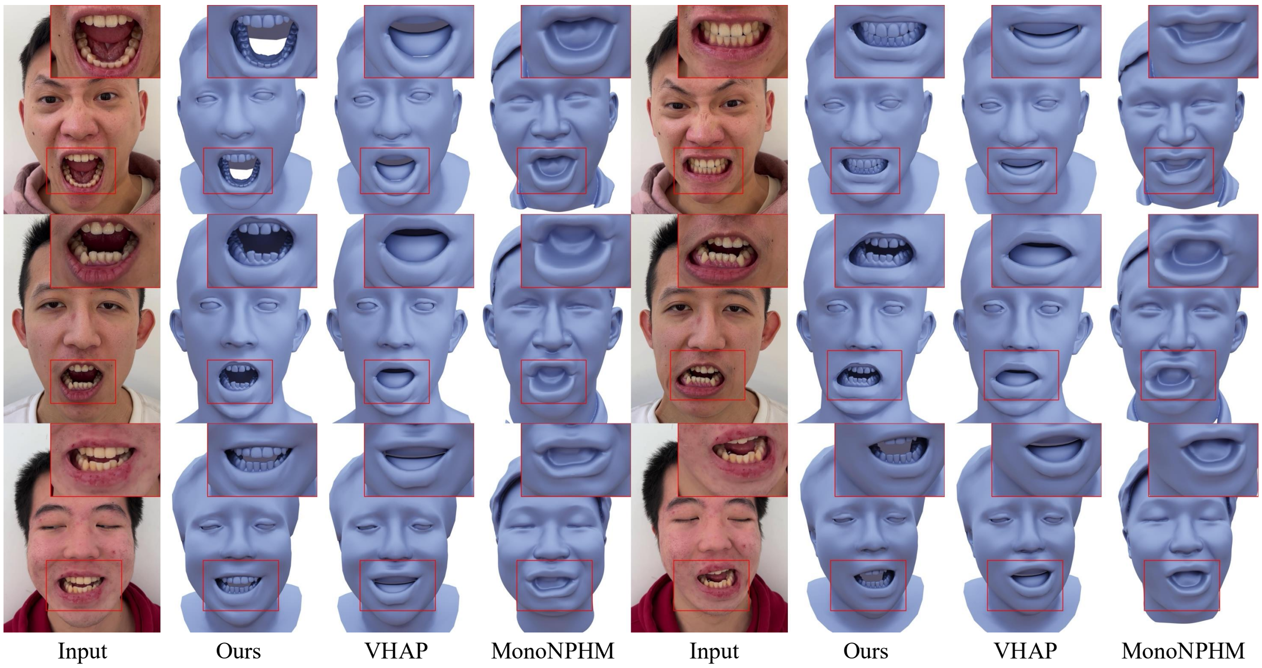

Figure 3. Comparison on monocular facial performance capture.

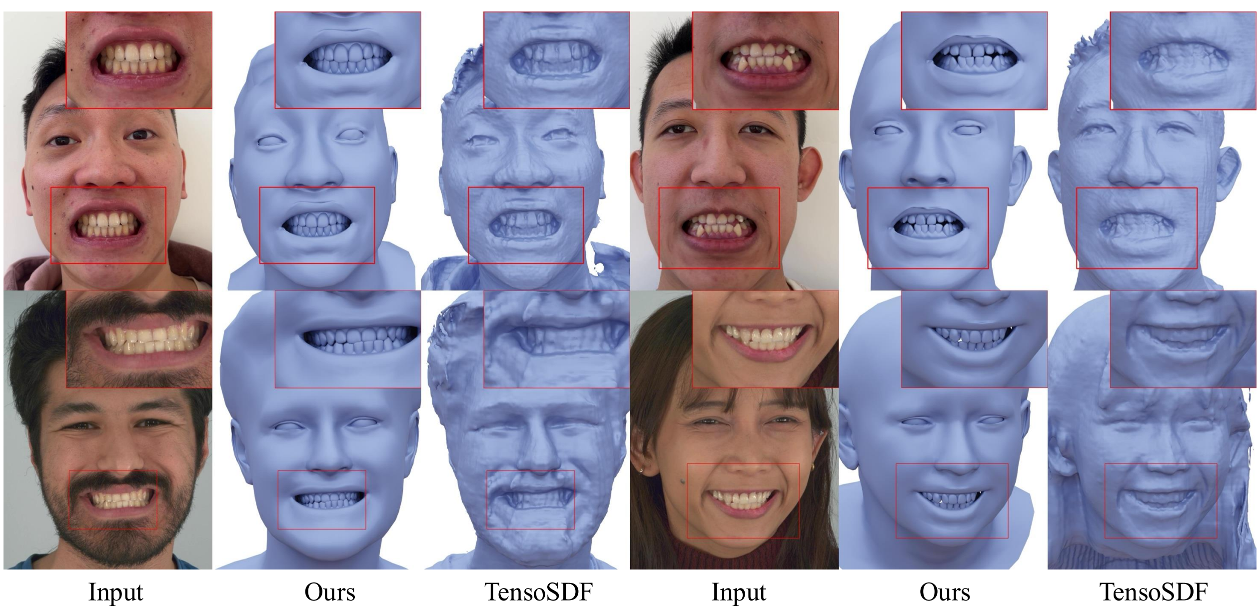

Figure 4. Comparison on shape reconstruction.

it struggles to reconstruct the details of the teeth, exhibiting lower quality. Our method can reconstruct the teeth shape more accurately because it leverages the neural SDF-based dental morphable model [61] to utilize teeth-specific prior, and employ the semantic mask-specific rendering loss to improve teeth alignment.

For quantitaive comparison, we also compute the geometric error metric following Sec. 5.2 for each input view on the synthetic dataset. The average metrics for each view at the region of the visible teeth are shown in Table 1. Our method quantitatively exceeds TensoSDF in teeth shape reconstruction.

5.4. Ablation study

SemanticXY loss. We show that the SemanticXY loss reduces sensitivity to the initial pose in the optimization process, as demonstrated by the ablation experiment removing the SemanticXY loss. As shown in Fig. 5, the baseline fails to accurate match the teeth's motion. In contrast, given the

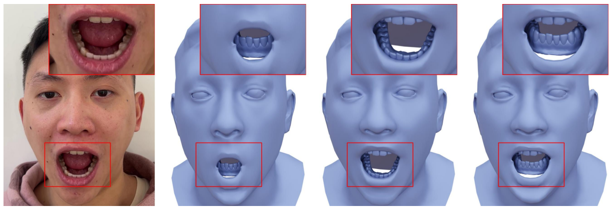

Figure 5. Ablation of the SemanticXY loss.

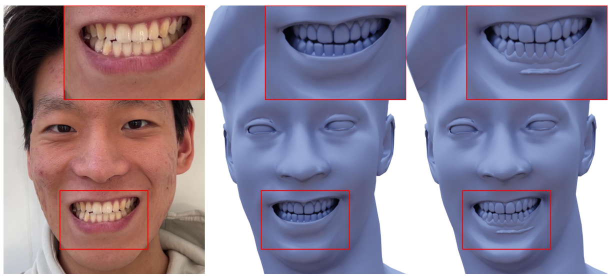

Figure 6. Ablation of the contact loss.

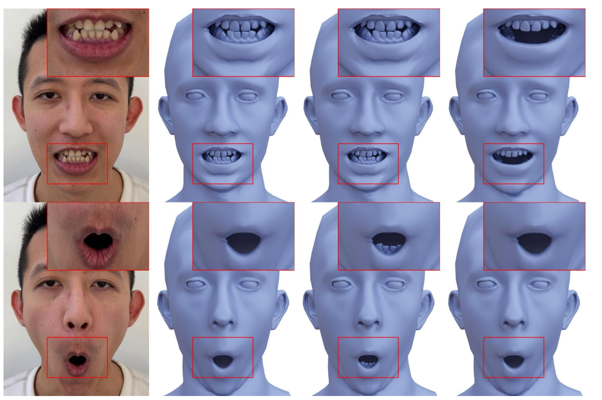

Figure 7. Ablation of the occlusion-aware optimization stategies.

same pose initialization, the method with SemanticXY loss exhibits more robust convergence behavior due to the global and long-range pixel correspondences.

Contact loss. The contact loss in Eq. (9) enables us to detect and prevent the penetration. As shown in Fig. 6, without the contact loss, the teeth and the face intersect, resulting in artifacts.

Occlusion-aware optimization stategies. As shown in Fig. 7, without the face-vanishing strategy and the teeth-hiding strategy, the tracking of teeth tends to get stuck in local minima under facial expressions with frequent occlusion state transitions, especially at the switching between the fully occluded and partially visible states.

Quantitative ablation results are shown in Tab. 2. Since Chamfer Distance can only be calculated on frames with visible teeth, this metric inherently cannot evaluate our occlusion-aware optimization stategies. To provide a more comprehensive assessment, we also report teeth mask mIOU across all frames to evaluate teeth alignment.

Reconstruction		Performance Capture	
Method	CD↓(10^{-5})	Method	CD↓(10^{-5})
Ours	**1.5674**	Ours	**2.8830**
TensoSDF	2.0312	VHAP	9.1174
–	–	MonoNPHM	5.9791

Table 1. Quantitative comparison on shape reconstruction and performance capture. **Bold** indicates the best results.

	CD↓(10^{-5})	mIoU ↑
Ours	<u>2.8830</u>	**0.7926**
w/o. semanticXY loss	8.7228	0.5872
w/o. contact loss	2.9165	<u>0.7915</u>
w/o. face vanishing	4.4268	0.7295
w/o. teeth hiding	**2.8363**	0.7621

Table 2. Quantitative ablation results. **Bold** indicates the best results and <u>underline</u> indicates the second best results.

6. Limitation

Our tracking method relies on accurate segmentation of the teeth and face. The current teeth segmentation we adopt uses the semi-automatic SAM2 [44]. Although this process requires users to label only 4-5 key frames per sequence (approximately 3 minutes total), it reduces the automation and ease of use of our tracking method. However, our method is not tied to any specific segmentation approach and can benefit from advances in teeth segmentation techniques, and recent advances in customizing SAM for class-specific detection (e.g. Personalize-SAM [63]) offer promising pathways to further automation. Additionally, improving the face reconstruction accuracy is an orthogonal direction, and our tracking method for teeth, lips, and face could be applied to any facial parametric model that contains anatomically meaningful skull and jaw. In the future, better face parametric models can benefit our method.

7. Conclusion

In conclusion, we have introduced the first method for personalized dental reconstruction and teeth-inclusive facial performance capture using only a smartphone camera. Our approach combines mesh and SDF representations, leverages semantic teeth segmentation for supervision, and addresses occlusion challenges through specialized optimization strategies. Experiments demonstrate that our method shows superior performance, produces accurate teeth geometry with precise motion tracking.

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